

Lab 3: Component Modelling & Architectural Pattern Selection

Smart Hostel Maintenance and Resource Management System

Architecture: Layered Architecture

1. System Overview

The Smart Hostel Maintenance and Resource Management System helps students, maintenance staff, and administrators manage hostel maintenance requests and resources through a structured software system. The architecture separates user interaction, business operations, notifications, and persistent data so that each component has a clear responsibility.

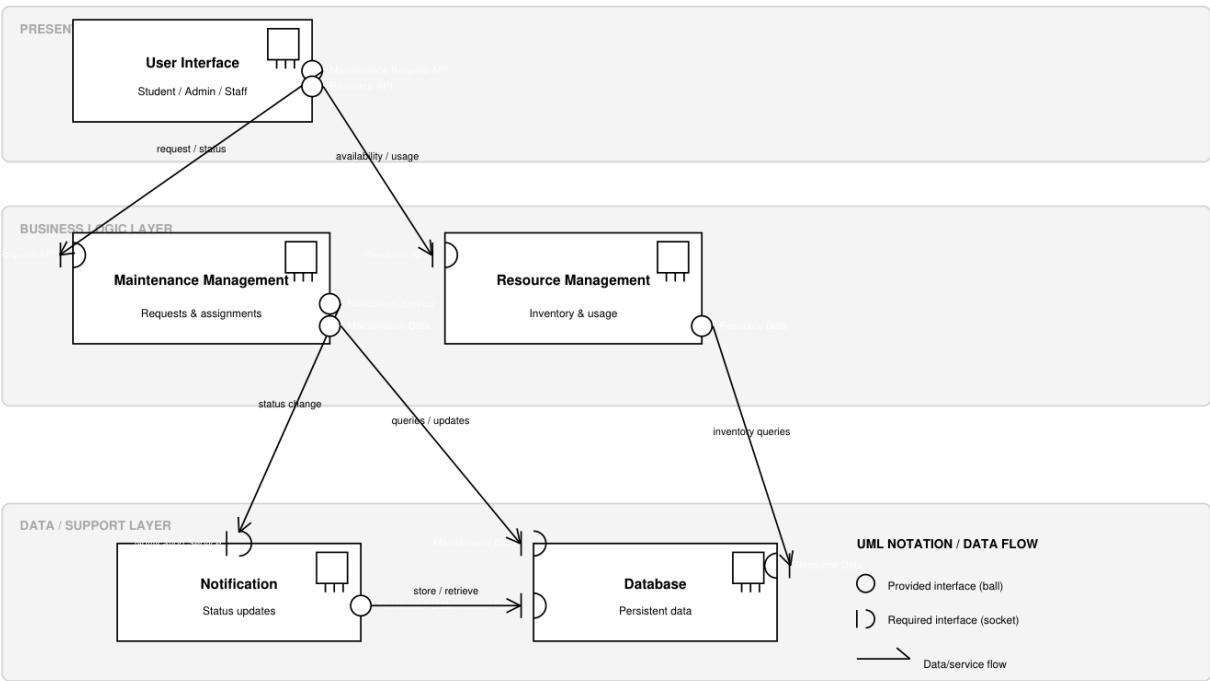
2. Main Components

Component	Responsibility
User Interface Component	Provides interfaces for students, maintenance staff, and administrators to submit requests, view information
Maintenance Management Component	Creates, updates, assigns, and tracks maintenance requests and repair activities.
Resource Management Component	Manages hostel resources, availability, inventory information, and resource usage.
Notification Component	Provides status-change and task-related notifications to relevant users.
Database Component	Stores student, staff, maintenance, resource, and notification data persistently.

3. UML Component Diagram

Smart Hostel Maintenance and Resource Management System

UML Component Diagram — Layered Architecture



Five components, six interfaces, UML component notation, labelled data/service flows, and component interactions.

The diagram uses UML component rectangles and shows provided interfaces using ball notation and required interfaces using socket notation. Labelled connections represent service requests, notifications, database queries, and data updates.

4. Interface and Data-Flow Details

Interface	Connected Components	Purpose
Maintenance Request API	User Interface ↔ Maintenance Management	Submit maintenance requests and view request/status information.
Resource API	User Interface ↔ Resource Management	Request resource availability information and record usage.
Notification Service	Maintenance Management ↔ Notification	Send status and assignment updates when maintenance activity changes.
Maintenance Data	Maintenance Management ↔ Database	Read and update maintenance request and assignment records.
Resource Data	Resource Management ↔ Database	Read and update resource and inventory records.
Notification Data	Notification ↔ Database	Store and retrieve notification information.

5. Architecture Selection & Written Justification

Architecture Selection: We chose **Layered Architecture** for the Smart Hostel Maintenance and Resource Management System.

Architectural Choice

The system is organized into a Presentation Layer, Business Logic Layer, and Data/Support Layer. The Presentation Layer handles user interaction, the Business Logic Layer manages hostel operations, and the Data/Support Layer provides persistent storage and notification support.

Reason 1 — Separation of Concerns

Layered architecture separates user interaction, business operations, and data management. Each component has a focused responsibility, making the system easier to understand and maintain.

Reason 2 — Maintainability

Maintenance management, resource management, and notifications are represented as separate components. Individual functionality can therefore be modified or extended without unnecessarily changing unrelated parts of the system.

Security Advantage

The layered structure can restrict direct access to the database. Requests can pass through the business layer, where validation and authorization checks can be applied before database operations are performed.

Performance Benefit

Separating business operations from data storage allows requests to be handled by the appropriate component and avoids requiring every user interaction to directly access the database. This can reduce unnecessary database operations and improve system responsiveness.

6. Component Interaction Flow

Typical maintenance flow: A student submits a maintenance request through the User Interface. The Maintenance Management Component processes the request, stores the relevant information in the Database, and communicates with the Notification Component when a status or assignment update needs to be communicated. Resource-related requests are handled by the Resource Management Component, which uses the Database for resource and inventory information.

Overall design benefit: Separating user-facing functionality from core hostel operations and persistent data creates clear component responsibilities and dependencies, making the system easier to maintain and extend.